

Visvesvaraya National Institute of Technology, Nagpur

**Electronics and Communication Engineering Department
Executive M.Tech. in Applied AI**

Cohort 3A, 3B sem. 3, Summer 2026

Eval. 3 (10/05/2026)

Detection of AI Generated Product Reviews

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[Click here to open Github repo](https://github.com/nikhilgbharadwaj/Detecting-AI-Generated-Product-Reviews-)

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Problem Statement

- AI-generated fake reviews are flooding e-commerce platforms, distorting ratings and misleading buyers.
- Existing detection methods are opaque, generator-specific, and dependent on paid infrastructure.
- This project builds a hybrid, explainable, locally-deployable detection system. The system transparently communicates the reasoning behind each verdict to the end user.

Dataset Snapshot original (~40k reviews): theArijitDas/Fake-Reviews-Dataset/

category string · classes	rating float64	text string · lengths	label int64
Home_and_K_10%	33.49.4%	2438564.8%	150.1%
Home_and_Kitchen		5 Love this! Well made, sturdy, and very comfortable. I love it!Very pretty	1
Home_and_Kitchen		5 love it, a great upgrade from the original. I've had mine for a couple of years	1
Home_and_Kitchen		5 This pillow saved my back. I love the look and feel of this pillow.	1
Home_and_Kitchen		1 Missing information on how to use it, but it is a great product for the price! I	1
Home_and_Kitchen		5 Very nice set. Good quality. We have had the set for two months now and have not been	1
Home_and_Kitchen		3 I WANTED DIFFERENT FLAVORS BUT THEY ARE NOT.	1
Home_and_Kitchen		5 They are the perfect touch for me and the only thing I wish they had a little more space.	1
Home_and_Kitchen		3 These done fit well and look great. I love the smoothness of the edges and the extra	1
Home_and_Kitchen		5 Great big numbers & easy to read, the only thing I didn't like is the size of the	1
Home_and_Kitchen		5 My son loves this comforter and it is very well made. We also have a baby	1
Home_and_Kitchen		5 As advertised. 5th one I've had. The only problem is that it's not really a	1
Home_and_Kitchen		5 Very handy for one of my kids and the tools are included in the package. I have one in	1
Home_and_Kitchen		5 Did someone say, "Oriental for \$60"? It is a great product for the	1
Home_and_Kitchen		1 These are so flimsy! They are not the quality you would expect from a piece of furniture.	1
Home_and_Kitchen		5 Makes may tea with out stirring. The only problem is that it's kind of hard to put	1
Home_and_Kitchen		5 Absolutely adorable! And excellent price. We have had the wooden ones for a few months now and they	1
Home_and_Kitchen		5 Love this! Perfect size for an entire family!Very good quality.	1
Home_and_Kitchen		5 These look beautiful and so nice. The only problem is that it's not really a mesh one.	1
Home_and_Kitchen		5 Exactly what you would expect. I love the look and feel of this pillow.	1
Home_and_Kitchen		5 10 Stars, I would highly recommend this item. We love this blanket.	1
Home_and_Kitchen		5 Great little egg masher. We have had it for a few months now and it's been	1
Home_and_Kitchen		5 As advertised. Easy to use. Love the colors. Also, the dimensions are just	1
Home_and_Kitchen		5 it is exactly as pictured. I love the look and feel of this pillow.	1
Home_and_Kitchen		5 Fantastic, just what I wanted!!! I love the look and feel of this pillow.	1

Dataset Snapshot New: (Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B, Phi-3 Mini) (~46k reviews)

category string	rating float64	text string	label int64
Kindle_Store	3	Argh, this book is very frustrating. It's sort of like reading a stream of consciousness with, lots of quick random changes of topic. Sort of like giving a character an accent it's sorta interesting, but it got old quickly. It's also a thriller, and it uses a bunch of cheap tricks to achieve that. Every other page is ...	0
Office_Products	5	5/5 stars! I'm blown away by the Office_Products item I purchased! As a tech enthusiast, I was impressed by its sleek design, intuitive interface, and top-notch performance. The quality is exceptional and it's already become an essential tool in my daily workflow. Highly recommend for anyone in the tech space!	1
Electronics	5	Now you can mount 2 speakers on top of each other. The only reason I ordered this is because I had a friend who was	1
Pet_Supplies	5	I have two small breed dogs and I thought it would be a nice addition to the litter box. The top is a nice bright color and is easy to clean. I used this to clean my older cat's litter box, and it did not come with a large filter.	1
Office_Products	5	5/5 Stars** I recently purchased the Office Products "Ultimate Notebook Set" and I'm thoroughly impressed. The set includes a high-quality notebook, a pack of colorful pens, and a set of sticky notes. The notebook itself is beautifully designed with a sturdy cover and smooth paper. The pens are vibrant and write smoot...	1
Electronics	5	Bought the adapter to go with my extra charging cable. Worked well with Samsung cable.	0
Pet_Supplies	5	My Girls just loved this toy!!!! Took them awhile to get it, but they figured it out :)	0
Movies_and_TV	5	Fun, intelligent, fascinating historical drama. In a class by itself in a world of "so much the same" contemporary crime dramas!	0
Books	4	A very good and interesting book. The characters are well developed and believable. The writing is good.I read this book in the middle of the night, and it was great. The story was well told	1
Sports_and_Outdoors	5	This is a great leash, the accordion style is much better than the one that came with my SUP board-that one got in the way I tried my husbands leash like this and am much happier	0
Tools_and_Home_Improvement	5	Does it's Job. But you know what I mean? I've never used this for a job	1
Pet_Supplies	5	My dogs and cats like this food and the taste is good. My dogs have been eating this	1
Pet_Supplies	4	A bit more light weight, but the design is just too large. The only downside is	1
Pet_Supplies	5	I purchased two of these and both work great. I will keep them in the house and use them when we go to bed. The first one was a little too big for my cat and it didn't fit. The second one was too small for my dog and it was too small for my dog. I will reorder these again. I will definitely buy another one.This is...	1
Toys_and_Games	3	The cards (with cats) are a little too long. My husband and I have been wanting to get a bunch of the cards,	1
Tools_and_Home_Improvement	5	Easy to use and works great. The only problem is that it comes with a small screw driver.	1
Home_and_Kitchen	5	These are great. I used to keep all my siverplate, serving pieces, and napkin rings in a silverware box that was so big it had to be put in a closet where it quickly got stacked with "stuff" or put up too high to get without a stepstool. It seemed more bother than it was worth to use. Now all my pieces are close by an...	0
Tools_and_Home_Improvement	5	Nice simple kit I can use during my computer builds.	0
Books	3	I know that these books are an easy read, and I will admit that I have been tempted to finish them. The first half of the book is a bit predictable, and it's worth it. It's a fairly short story about two women with an interesting relationship.	1
Books	5	Spiritual. I would recommend this book for the spiritual journey it takes you through out the story. It is a feel good book.	0
Electronics	5	Super easy upgrade. I was going to help my kid do as I have an IT background but he read the directions and did it himself.	0

Dataset analysis of the reviews: 46,411 labelled reviews

Splits and Label Balance

Split	Rows	Human	AI	AI %
train	35,972	16,156	19,816	55.1%
validation	4,496	2,053	2,443	54.3%
test	4,497	2,023	2,474	55.0%
heldout	1,446	0	1,446	100.0%
Total	46,411	20,232	26,179	56.4%

Review Length

Class	Mean	Median	Std Dev	Min	Max
Human	73.6	42.0	76.1	5	373
AI-generated	67.6	39.0	64.4	1	318

Star-Rating Distribution

Class	1★	2★	3★	4★	5★
Human	5.4%	5.0%	9.1%	20.0%	60.5%
AI-generated	6.0%	5.7%	9.3%	20.7%	58.3%

Lexical and Vocabulary Statistics

Metric	Human (mean)	AI (mean)	Difference
Type-token ratio (vocabulary diversity)	0.809	0.742	-0.067
Average word length	4.204	4.140	-0.064
Unique words per review	50.82	41.59	-9.23

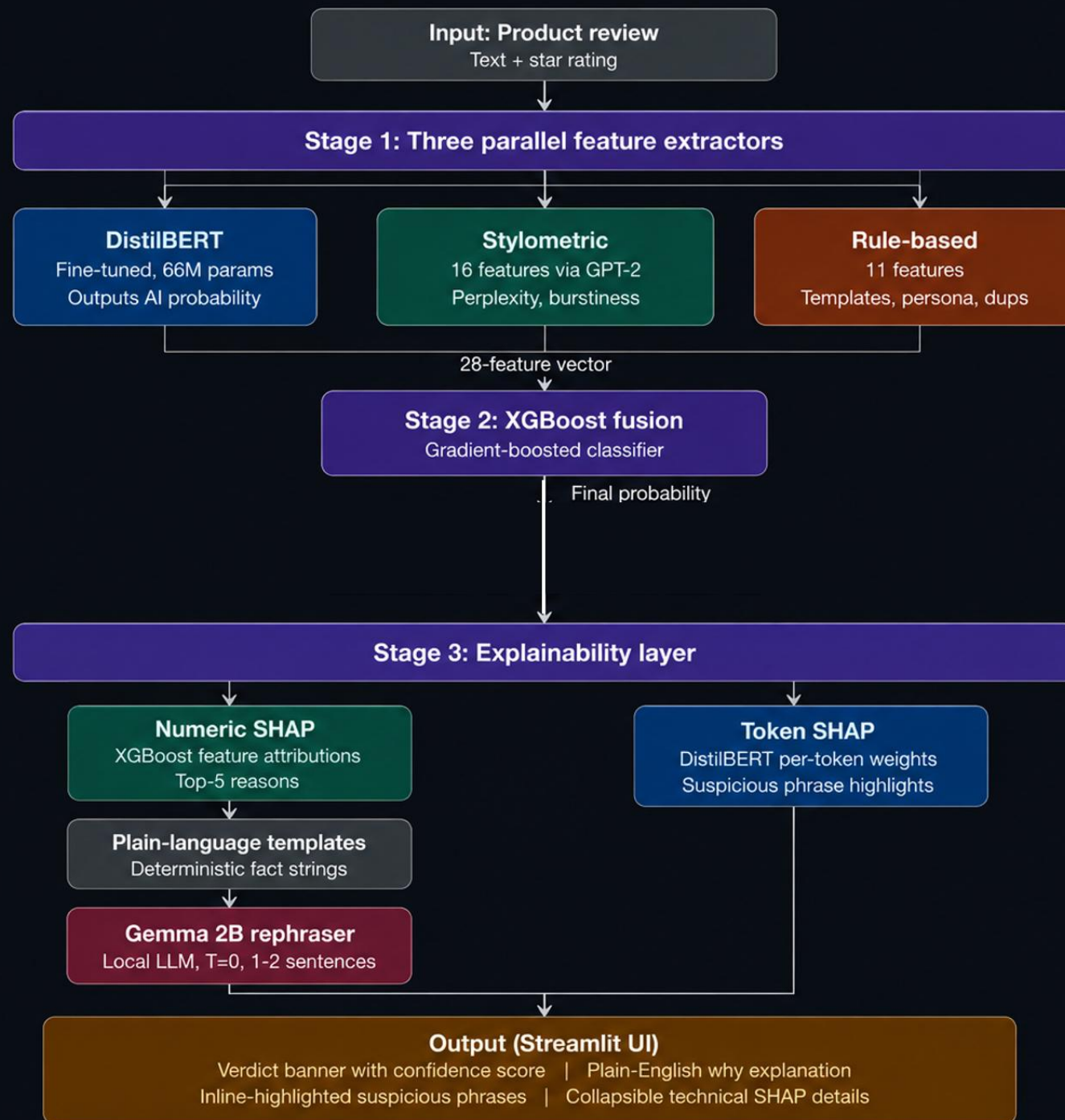
Word Usage Differences

Words over-represented in AI	Words over-represented in Humans
changer, exceptional, exceeded, sleek, durability	here, etc, along, believe, went
materials, expectations, thrilled, top-notch	actually, maybe, though, until, isn't
busy, persona-based vocabulary, marketing terms	casual contractions, time anchors, hedging

Common Bigram Patterns

AI top bigrams	Human top bigrams
would recommend, highly recommend	year old, well made, works great
recommend anyone, well written	better than, can't wait, don't know
only problem, recently purchased	set up, really like, long time
honest review, will keep	good quality, even though, buy again

Model Architecture:



"perplexity": Statistically too predictable
(perplexity {value} for typical human reviews)

Stylometric

#	Feature name	What it measures
1	perplexity	How predictable the entire review is, scored by GPT-2. Lower = more AI-like.
2	mean_sentence_perplexity	Average per-sentence perplexity.
3	burstiness	Standard deviation of perplexity across sentences. Lower = more uniform = more AI-like.
4	type_token_ratio	Unique words ÷ total words. Vocabulary diversity.
5	mean_sentence_length	Average words per sentence.
6	std_sentence_length	Variance of sentence length within the review.
7	mean_word_length	Average characters per word.
8	punctuation_density	Punctuation characters ÷ total characters.
9	exclamation_ratio	Number of '!' ÷ word count.
10	question_ratio	Number of '?' ÷ word count.
11	caps_ratio	All-uppercase words ÷ total words.
12	generic_adjective_ratio	Frequency of words like 'great', 'amazing', 'perfect'.
13	hedging_ratio	Frequency of words like 'perhaps', 'quite', 'rather'.
14	first_person_ratio	Frequency of 'I', 'my', 'me', 'mine'.
15	digit_ratio	Digits ÷ total characters.
16	word_count	Total number of words.

Rule based

#	Feature name	What it measures
1	rule_template_hits	Count of known LLM template phrases hit (e.g., "I was blown away by", "highly recommend", "sleek design").
2	rule_buyer_persona_phrases	Count of buyer-persona phrases (e.g., "as a busy parent", "as a tech enthusiast").
3	rule_starts_with_rating	Boolean (0 or 1): does the review open with "5/5 stars", "★★★★★", or similar?
4	rule_markdown_artifacts	Count of leftover markdown formatting (bold markers, headers, bullet markers from LLM output).
5	rule_pros_cons_structure	Boolean: does the review use a "Pros: ... Cons: ..." section structure?
6	rule_superlative_density	Density of clusters of generic adjectives close together.
7	rule_sentiment_rating_mismatch	Boolean: does the sentiment of the text contradict the star rating?
8	rule_emoji_count	Number of emoji characters in the review.
9	rule_avg_paragraph_perfection	How uniform paragraph lengths are. Lower coefficient of variation = more uniform = more AI-like.
10	rule_near_duplicate	Boolean: is this review a near-duplicate of another review in the corpus, found via MinHash + LSH?
11	rule_max_jaccard	Maximum Jaccard similarity (using 5-shingle MinHash) to any other review in the corpus.

Model Performance Results

DistilBERT Training Curve

Epoch	Train Loss	Val Loss	Val Acc	Val Prec	Val Recall	Val F1	Val False Positive Ratio
1	0.0972	0.2007	0.9242	0.8855	0.9881	0.9340	0.1520
2	0.0523	0.1004	0.9655	0.9490	0.9898	0.9689	0.0633
3	0.0254	0.1087	0.9686	0.9546	0.9894	0.9717	0.0560

DistilBERT Test-Set Performance

Set	Accuracy	Precision	Recall	F1	False Positive Ratio
Validation	0.9686	0.9546	0.9894	0.9717	0.0560
Test (in-distribution)	0.9706	0.9574	0.9907	0.9738	0.0539
Held-out generator (Qwen 2.5)	0.9550	1.0000	0.9550	0.9770	0.0000

Hybrid Fusion Performance

Configuration	Accuracy	Precision	Recall	F1	False Positive Ratio
DistilBERT alone	0.9700	0.9570	0.9899	0.9732	0.0544
Fusion (default)	0.9722	0.9730	0.9766	0.9748	0.0331

DEMO

Conclusion

- Developed a hybrid AI review detector combining DistilBERT, stylometric analysis, rule-based heuristics, and XGBoost.
- Achieved 97.26% test accuracy, 0.9752 F1-score, and only 3.31% false positive rate.
- Reduced false positives by 39% compared to standalone DistilBERT.
- Demonstrated strong cross-generator generalization with 95.5% recall on unseen Qwen 2.5 7B reviews.
- Integrated explainable AI using SHAP, fact-based reasoning, and a local LLM for plain-English explanations.
- Entire system runs offline on a single consumer GPU using free, open-source tools with a local Streamlit interface.